



LEITH RIGBY

Mechanical Engineering Student

PROFILE

Mechanical Engineering student at UBC Okanagan who likes taking ideas from concept to working systems. I pride myself on discipline, hard work, and pushing myself to new limits. Competitive team sports and two seasons crewing a Coast Guard rescue boat have taught me to communicate clearly, stay calm under pressure, and make the best of my time. My day starts early at the gym so the rest of it goes to studies and building.

CONTACT

PHONE:

250-258-3539

EMAIL:

leithr@telus.net

lrigby01@student.ubc.ca

LINKEDIN:

linkedin.com/in/leith

LOCATION:

Kelowna, British Columbia

SKILLS

Software

- SOLIDWORKS (CSWP, CSWA)
- Python
- C++
- Arduino

Fabrication

- 3D printing
- Laser engraving
- Iterative prototyping
- Electronics & wiring

Languages

- English
- French (DELFB2)

EDUCATION

University of British Columbia Okanagan

Bachelor of Applied Science – Mechanical Engineering

September 2024 – Present

Kelowna Secondary School

September 2021 – June 2024

- Double Dogwood (English/French)
- 4 AP, university-level courses
- DELFB2 French proficiency exam
- Dual Credit UBCO APSC 169 – Sustainable Engineering; First Place in the Design Process

TECHNICAL WORK EXPERIENCE

Canadian Coast Guard – Search and Rescue

Crew Member, Inshore Rescue Boat · Victoria, BC

May 2025 – Present (May – August 2025 & May – Sept 2026)

- Crewed a three-person Inshore Rescue Boat station (remote coastal locations) responding to marine search and rescue taskings
- Operated the Fast Rescue Craft (Zodiac) in remote postings where our crew was often the only resource available to persons injured or in distress (SVOP, SCT, RHIOT)
- First responder to marine and medical emergencies; delivered on-scene patient care and patient transfer to EHS (Marine Advanced First Aid, Standard First Aid & CPR-C)
- Completed hundreds of hours of SAR training to keep the crew ready for taskings on short notice
- Awarded Rookie of the Year for the Inshore Rescue Boat program, 2025 season

ENGINEERING DESIGN TEAMS

UBC Okanagan Marine Robotics (RoboSub)

Mechanical Lead, · Kelowna, BC

September 2025 – Present

- Oversee 8 sub-teams, totaling around 25 engineering students working on separate, complex projects.
- Building an Autonomous Underwater Vehicle (AUV) with the student design team for annual international competition
- Took ownership of the item dropper mechanism, running it from concept through design, prototyping and manufacturing
- Designed and prototyped a high-speed auxiliary AUV to complement the main vehicle

CERTIFICATIONS

- Certified SOLIDWORKS Professional (CSWP), Sep 2026
ID: C-HNUJG86K5F
- Certified SOLIDWORKS Associate (CSWA), Sep 2026
ID: C-LRNHX7Q54K
- Marine Advanced First Aid
- Standard First Aid & CPR-C
- Small Vessel Operator Proficiency (SVOP)
- SCT
- Rigid Hull Inflatable Operator Training (RHIOT)
- Restricted Operator Certificate – Maritime (ROC-M)
- Pleasure Craft Operator Card (PCOC)
- PADI Open Water Diver
- DELF B2 (French)

HOBBIES

- Weightlifting/ Health
- 3D Printing
- Engineering/ Designing
- Boating & marine operations
- Fishing

SPORTS

- Ice Hockey (competitive)
- Judo
- Soccer
- Baseball
- Watersport (Wakeboarding, Surfing)
- Scuba Diving

TECHNICAL PROJECTS

Handheld Night Vision Goggles

Designed and built functioning night-vision goggles using active infrared imaging, combining existing parts with new 3D printed components.

Two-Person Go-Kart

Designed, prototyped and iterated a large two-person go-kart from scratch. Began as a wooden, gravity-powered model and developed into a several-hundred-pound, gas-powered, fully roll-caged kart, all on a small budget.

Pneumatic Potato Cannon

Designed and built a pneumatic potato cannon from sprinkler parts, a DeWalt battery and 3D printed components.

Physics Demonstration Video – Fluid dynamics

Created a YouTube video for the Junior Breakthrough Challenge. Demonstrated the importance of the Bernoulli principle with a self-made fire display.

Hidden Bookshelf Door

Designed and built a hidden bookshelf that opens on an RFID tag, driven by a linear actuator and controlled by an Arduino.

OTHER WORK EXPERIENCE

1-800-GOT-JUNK? – Truck Team Member

Kelowna, BC

June 2024 – September 2024

- Loaded and organized the truck on customer sites while keeping the job moving and the site clean
- Communicated with customers, confirmed bookings and made on-site sales, quoting work to give the best value
- Handled payment collection

Parlour Ice Cream – Point of Sale Associate

Kelowna, BC

May 2021 – September 2023

- Handled customer sales and cash register operation through high-volume summer service
- Ensured an enjoyable customer experience from order to handoff
- Kept a clean, organized workspace

ACHIEVEMENTS AND AWARDS

- Rookie of the Year – Canadian Coast Guard, Inshore Rescue Boat (2025)
- First Place in the Design Process – APSC 169 (UBCO Engineering)
- Most Sportsmanlike Player (Hockey)